T3 Weinberg-operator coefficient comparison

Model	Ordered coefficient extracted from Matchete	Symmetric physical Wilson coefficient
T3-A	$A_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$
T3-A	$A_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$
T3-A	$A_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$
T3-A	$A_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$
T3-A	$A_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$
T3-A	$A_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$
T3-B	$A_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$
T3-B	$A_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$
T3-B	$A_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$
T3-B	$A_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$
T3-B	$A_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$
T3-B	$A_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\sqrt{3}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{S2}^2\lambda_{T3}M_{S1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_F^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S1}^2}\right)M_{S2}^2M_{S1}^2+\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_{S2}^2M_{S1}^2-\log\left(\frac{M_F^2}{M_{S2}^2}\right)M_F^2M_{S2}^2\right)}\right.$

Model	Ordered coefficient extracted from Matchete	Symmetric physical Wilson coefficient
T3-C	$A_5^{ij} =$ $-\frac{2}{3}\left(\frac{\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{\mathrm{S}2}^2\lambda_{\mathrm{T}3}M_{\mathrm{S}1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{\mathrm{S}1}^2}\right)M_F^2M_{\mathrm{S}1}^2-\log\left(\frac{M_F^2}{M_{\mathrm{S}1}^2}\right)M_{\mathrm{S}2}^2M_{\mathrm{S}1}^2+\log\left(\frac{M_F^2}{M_{\mathrm{S}2}^2}\right)M_{\mathrm{S}2}^2M_{\mathrm{S}1}^2-\log\left(\frac{M_F^2}{M_{\mathrm{S}2}^2}\right)M_F^2M_{\mathrm{S}2}^2\right)}\right.$	$C_5^{ij} =$ $\frac{2}{3}\left(\frac{\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{\mathrm{S}2}^2\lambda_{\mathrm{T}3}M_{\mathrm{S}1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{\mathrm{S}1}^2}\right)M_F^2M_{\mathrm{S}1}^2-\log\left(\frac{M_F^2}{M_{\mathrm{S}1}^2}\right)M_{\mathrm{S}2}^2M_{\mathrm{S}1}^2+\log\left(\frac{M_F^2}{M_{\mathrm{S}2}^2}\right)M_{\mathrm{S}2}^2M_{\mathrm{S}1}^2-\log\left(\frac{M_F^2}{M_{\mathrm{S}2}^2}\right)M_F^2M_{\mathrm{S}2}^2\right)}\right.$

Model	Ordered coefficient extracted from Matchete	Symmetric physical Wilson coefficient
T3-E	$A_5^{ij} = -\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{\mathrm{S}2}^2\lambda_{\mathrm{T}3}M_{\mathrm{S}1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{\mathrm{S}1}^2}\right)M_F^2M_{\mathrm{S}1}^2-\log\left(\frac{M_F^2}{M_{\mathrm{S}1}^2}\right)M_{\mathrm{S}2}^2M_{\mathrm{S}1}^2+\log\left(\frac{M_F^2}{M_{\mathrm{S}2}^2}\right)M_{\mathrm{S}2}^2M_{\mathrm{S}1}^2-\log\left(\frac{M_F^2}{M_{\mathrm{S}2}^2}\right)M_F^2M_{\mathrm{S}2}^2\right)}\right.$	$C_5^{ij} = \frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}M_{\mathrm{S}2}^2\lambda_{\mathrm{T}3}M_{\mathrm{S}1}^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{M_{\mathrm{S}1}^2}\right)M_F^2M_{\mathrm{S}1}^2-\log\left(\frac{M_F^2}{M_{\mathrm{S}1}^2}\right)M_{\mathrm{S}2}^2M_{\mathrm{S}1}^2+\log\left(\frac{M_F^2}{M_{\mathrm{S}2}^2}\right)M_{\mathrm{S}2}^2M_{\mathrm{S}1}^2-\log\left(\frac{M_F^2}{M_{\mathrm{S}2}^2}\right)M_F^2M_{\mathrm{S}2}^2\right)}\right.$